

graphically?

Author: Anyway, a graph for $s(t)$ always exists. The only "inconvenience" in your definition is that it is necessary to take into account the scale of units on the coordinate axes. If the unit of time (on the t -axis) and the unit of length (on the s -axis) are represented by segments of identical length, the instantaneous velocity at time t is

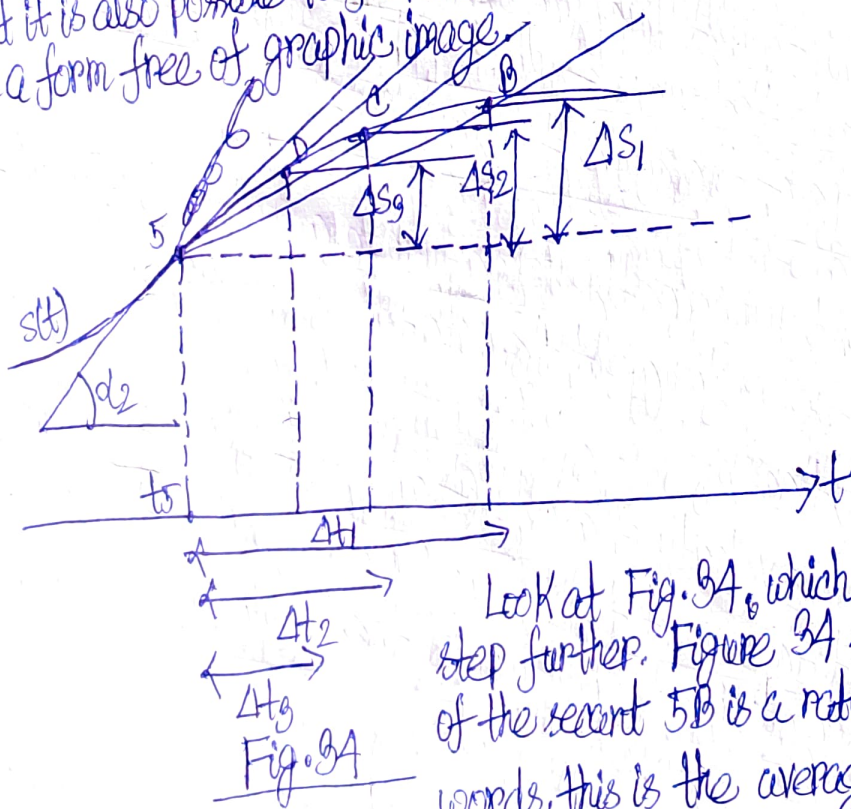
$$v_{\text{inst}}(t) = \frac{\text{unit of length}}{\text{unit of time}}$$

If, however, the segment representing one unit of length is n times greater than the segment representing one unit of time, the instantaneous velocity is

$$\frac{1}{n} v_{\text{inst}}(t) = \frac{\text{unit of length}}{\text{unit of time}}$$

This "inconvenience", however, has no principal significance.

But it is also possible to formulate a definition of the instantaneous velocity in a form free of graphic image



Look at Fig. 34, which carries Fig. 33 one step further. Figure 34 shows that the slope of the secant 5B is a ratio $\frac{\Delta s_1}{\Delta t_1}$. In other words, this is the average Δt_1 velocity for

the time interval from t_5 to $t_5 + \Delta t_1$. The slope of the secant 5C is $\frac{\Delta s_2}{\Delta t_2}$, that is, the average velocity for the time interval from t_5 to $t_5 + \Delta t_2$ ($\Delta t_2 < \Delta t_1$). The slope of the secant 5D is $\frac{\Delta s_3}{\Delta t_3}$, that is, the average velocity for the time interval